

Account types

Standard, Programmer, Administrator, Suspended and Group — what each one may do, and how to make one.

OpenQM gives every account the same VOC and leaves privilege to `SYSTEM` membership. SD Core does not. **An account is created into one of three tiers, and the tier decides what verbs its VOC contains.** There is a fourth tier, **suspended**, which is not a capability at all — it denies entry. Group accounts are a different thing again: a shared place, not a person.

The tier is no longer fixed at creation. `modify.account` moves an account between all four, in either direction, and rebuilds its VOC to match at once. See Changing an account afterwards.

The three capability tiers at a glance

	STANDARD	PROGRAMMER	ADMINISTRATOR
Verbs	81	81 + 42	81 + 42 + 20
VOC records on creation	354	396	416
Can run an application	yes	yes	yes
Can compile, catalogue, edit	no	yes	yes
Can create or configure files	no	yes	yes
Can administer accounts	no	no	yes
Windows <code>Administrators</code>	no	no	yes
Local console and Remote Desktop	denied	denied	allowed

`administrator` **implies** `programmer`. You do not need both keywords.

STANDARD — 82 VERBS

What an application needs and no more: query and list (`select`, `list`, `get.list` and family), spool and print, session and environment (`logto`, `date`, `who`, `set`), screen and message, prompt and input state, and eight read-only inspectors — `search`, `list.diff`, `list.item`, `list.common`, `list.vars`, `report.src`, `report.style`, `format`.

Everything an application built on SD invokes, and nothing that edits code or data in bulk.

PROGRAMMER — 42 MORE

The development set: the compilers, the two full-screen editors, the cataloguer, the file and index definition verbs, the bulk record editors and the process introspection verbs. See Programmer commands for what each one is for.

THE COUNTS ARE ARITHMETIC, NOT OBSERVATION

Installed `NEWVOC` holds 395 names, of which `%t` is a dynamic-file artefact and the two tier lists are never copied — so **392 records reach a full VOC**. `create.account` then adds four of its own (`$command.stack`, `$hold`, `$savedlists`, `bp`):

ADMINISTRATOR	392 + 20 + 4 = 416
PROGRAMMER	392 + 4 = 396
STANDARD	392 - 42 + 4 = 354

A standard account's total did not move when `micro` was added, because `micro` joined `NEWVOC` and `TIER.OMIT.STANDARD` at once — it is on both sides of the subtraction.

If your counts differ, one of the two tier lists differs — which is worth reporting.

The same two numbers are what a tier change reports, so you can predict them: moving between standard and programmer is **42** records either way, and between programmer and administrator **20**. A change that reports a different number, or zero where it should have moved something, is worth reporting for the same reason.

ADMINISTRATOR — 20 MORE

Account and grant administration, system-wide state, and the shell escapes. See Administrator commands.

None of the three is a wall. An administrator can copy any verb into any account's VOC afterwards. **The reduced VOC is the posture an account starts in, not a boundary anything enforces.** The boundaries that are enforced are the operating system's file permissions, the ssh confinement, and the `os.users` permit list — not the contents of a VOC.

Suspended — the fourth tier, and the only one that is a wall

A suspended account cannot be entered. It is for an account that should stop working for a while — somebody on leave, a login being looked into — and it is refused at all three ways in:

ssh, or the console	Account FRED is suspended
<code>logto</code> from another account	Account FRED is suspended
the API	User not allowed in requested account

The API wording is deliberately the same one it gives for an account that does not exist and for one you are not granted, so the API cannot be used to find out which accounts exist or what state they are in.

It takes nothing away, which is why lifting it is free. The VOC is left exactly as it is, no Windows group membership moves, and the tier it displaced is remembered — so bringing the account back puts it exactly where it was, with nothing for you to write down. Suspending is not a substitute for deleting: it is reversible on purpose.

An elevated administrator can still `logto` into a suspended account. That is deliberate — looking at a suspended account is the usual reason to have one, and anybody elevated could lift the suspension anyway. **What a suspension denies is the account's own user.**

And a suspended administrator is still a Windows administrator. SD refuses them; Windows does not. They keep their `Administrators` membership and their `os.users` record, so they can still elevate on the machine and still reach any account they could reach before. **If you are suspending an account to contain somebody rather than to park it, suspend it in Windows too.**

Creating an account

```
create.account user <name> {administrator | programmer}
                        <ssh | api | both | none> {no.query}

create.account group <name> {no.query}

create.account other <name> <pathname> {no.query}
```

ONE OF `SSH`, `API`, `BOTH`, `NONE` IS REQUIRED

There is no default, on purpose. An account that should only ever be reached with `logto` says `none` and means it. The old silent behaviour — ssh yes, API no — could not tell that apart from somebody who had not thought about it.

An administrator account always gets both and needs no keyword. Group accounts take none of this: they have no Windows account.

WHAT CREATING A USER ACCOUNT ACTUALLY DOES

Makes a Windows local account	created disabled, then enabled when the password is set
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Creates the group <code>sdu_<name></code>	and writes it to the account record
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Joins <code>sdusers</code>	which is what grants access to the data tree
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Joins `sdsshonly` **unless** `administrator` — this is what denies the console and Remote Desktop

Joins `Administrators` **only** with the `administrator` keyword

Writes the tier to `ACCOUNTS` field 5 so `LOGIN` cannot undo it at the next update

Prompts for a password in SD, masked; it never goes on a command line

A user account cannot be created without a password. Refusing the prompt creates nothing at all. Previously it left an account you could not sign in to.

Elevation is not optional. Creating a Windows account needs an elevated token, and an ordinary SD session has a filtered one. Account creation works from the installer and from an elevated terminal, and not from a normal session.

`create.account user ... ssh` and `... both` **are refused when the machine has no ssh server**, with a warning saying why: the account would be denied the console and Remote Desktop and have no ssh to arrive on, so it could sign in nowhere. `api` and `none` still work. The test is made against the machine when you type the command, so installing an ssh server later makes `ssh` start working. See Installing SD Core.

Continued in

Managing accounts — group accounts, sharing an account, changing one afterwards, and deleting it.
